# DSCI-560
# Lecture 11: Examples of Data Wrangling
## Data Science Professional Practicum

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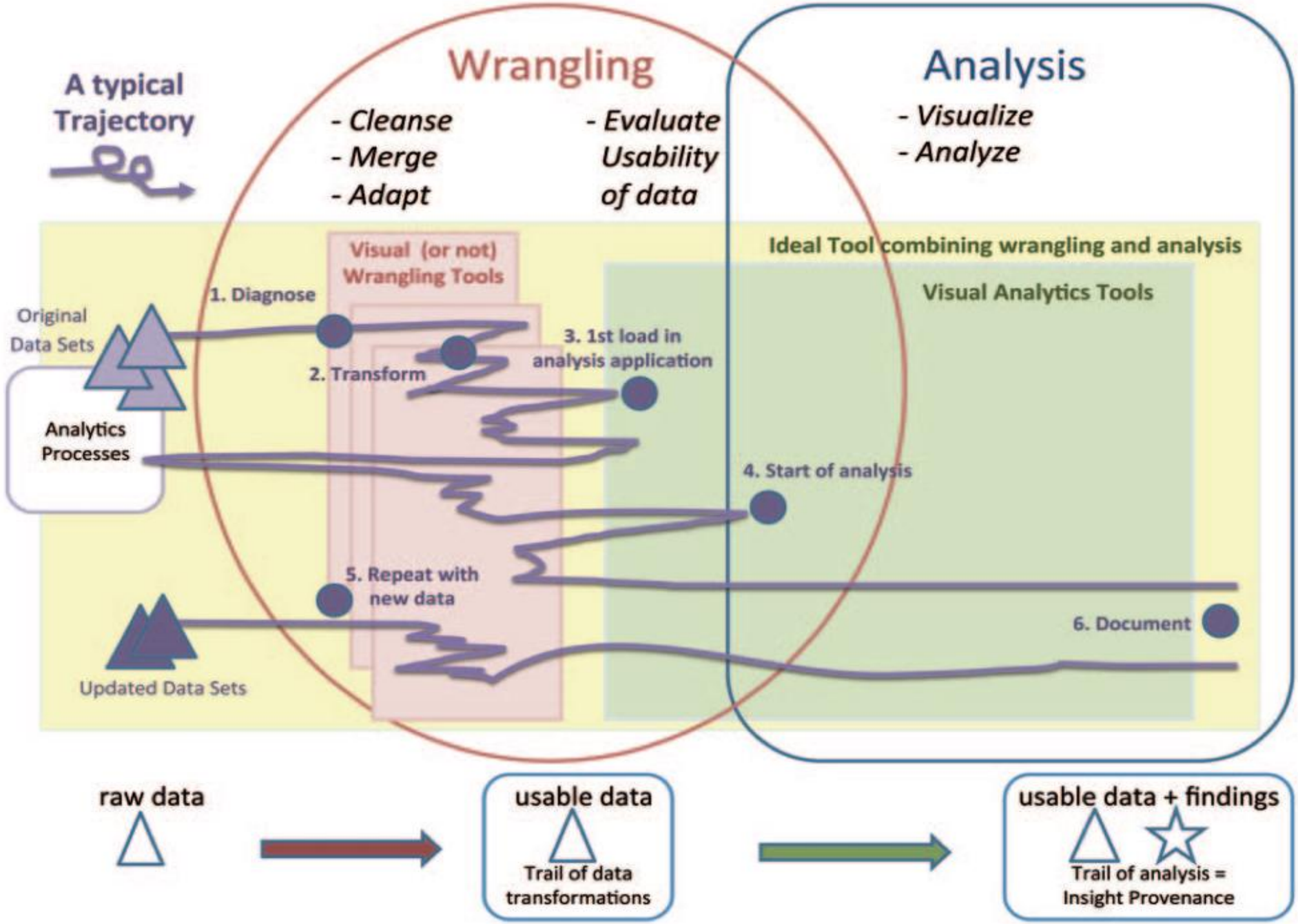

A typical Trajectory
Wrangling
- Cleanse
- Merge
- Adapt
- Evaluate Usability of data
Analysis
- Visualize
- Analyze
Ideal Tool combining wrangling and analysis
Visual (or not) Wrangling Tools
Visual Analytics Tools
Original Data Sets
Analytics Processes
1. Diagnose
2. Transform
3. 1st load in analysis application
4. Start of analysis
5. Repeat with new data
6. Document
Updated Data Sets
raw data
usable data
Trail of data transformations
usable data + findings
Trail of analysis = Insight Provenance

# Data Wrangling

Transforming and mapping "raw" data into a format suitable for some sort of analysis

- Highly Iterative
- Highly Task Dependent
- Time Consuming

https://en.wikipedia.org/wiki/Data_wrangling

# Typical Data Wrangling Steps

- **Discovering** - Explore the raw data to gain a better understanding of the data: different data is worked and organized in different ways.
- **Structuring** – Raw data is typically unorganized and much of it may not be useful for a particular analysis. Extract the useful parts and organize them (shape/sort/aggregate/…) for easier computation and analysis.
- **Cleaning** – Assuring data quality: Examples include converting all dates to a single format, handling outliers and missing data, removing/merging duplicates, etc.
- **Enriching** – Is additional data available that would enhance the current data set or planned analyses?
- **Validating** – More quality assurance: Similar to **structuring** and **cleaning**, use repetitive sequences of validation rules to assure data consistency as well as quality and security. An example of a validation rule is confirming the accuracy of fields by cross checking data.
- **Publishing** – Make data accessible for analysis (file format, location, etc.).

*Be sure to document any and all data wrangling steps and logic!!*

https://en.wikipedia.org/wiki/Data_wrangling

# Data Wrangling

- Get data into a structured form suitable for analysis
  - Variously called: data preparation, data munging, data curation
  - Also often called ETL (Extract-Transform-Load) process
- Often the step where majority of time (80-90%) is spent
- Key steps:
  - Scraping: extracting information from sources, e.g., webpages, spreadsheets
  - Data transformation: to get it into the right structure
  - Data integration: combine information from multiple sources
  - Information extraction: extracting structured information from unstructured/text sources
  - Data cleaning: remove inconsistencies/errors

# Data Munging

- Pretty much synonymous with data wrangling
- When they are used differently…
  - *Data wranging* seems to refer more to processes that start with very raw data
    - (e.g. data scraped from the www)
  - *Data munging* seems to refer to further processing already- or partially-wrangled data with an eye toward new analyses
    - (e.g. linking and further cleaning drivers license and taxpayer lists to count unique persons in the union of the lists, and estimate how many people are missing from both lists)

# Typical Data Munging Steps

- The core steps of data munging are the same as data wranging
  - Exploration (Discovering)
  - Transformation (Structuring)
  - Enrichment (Enriching)
  - Validation (Validating, including Cleaning)
- Like data wrangling, data munging is
  - Iterative
  - Highly task-dependent

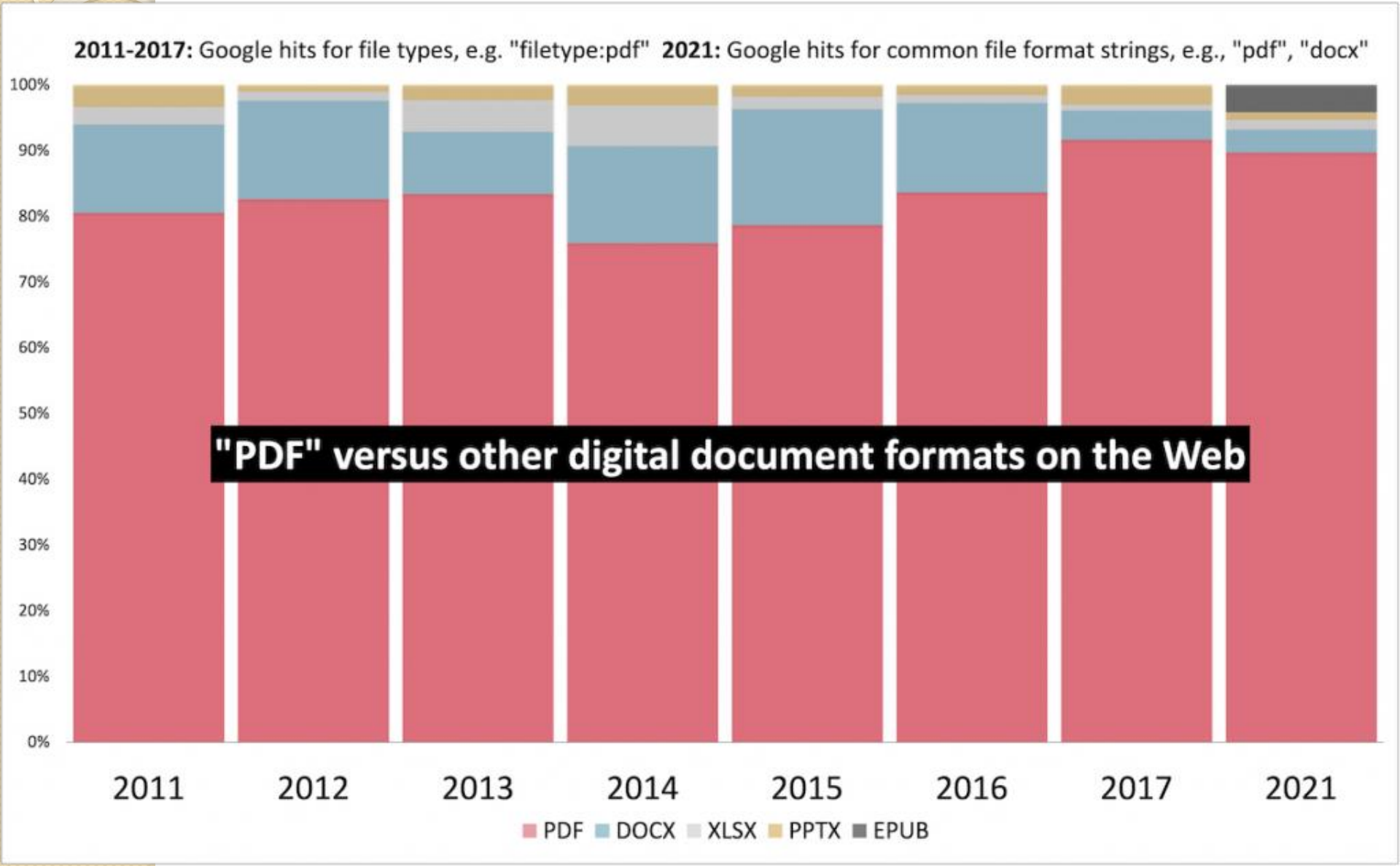
*Be sure to document any and all steps and logic in data munging!!*

https://www.talend.com/resources/what-is-data-munging/

# Example: Common Internet Documents

- Non-ASCII Doc
  - 90% PDF
  - 3% DOCX
  - 2% XLSX
  - 1% PPTX
  - 4% EPUB
- ASCII Text
  - HTML
  - XML
  - JSON
  - CSV
  - Unstructured

# Digital Documents on the Web



# Conversion

- Data Formats
  - Various into a common format
- Scanned Text into Texts
  - Optical Character Recognition
  - Higher level corrections
- Audio to Texts
  - Voice Recognition
  - Natural Language Processing
- Images to Texts
  - Image Recognition
- Multimedia
  - Combination of Images and Audio

# Data Examination

- Input Format
  - Image versus Text
    - Extract text and images
    - Convert images to text with Optical Character Recognition
    - Tag non-text images with text
  - Structured versus Unstructured
    - Existing APIs and Tools
    - Custom Parser
  - Application Relevant Data Selection
    - Keep relevant data
    - Filter out or archive the rest
- Output Format
  - Common format for all documents
  - Standard format supported in the backend

# Example: Methane Leakage Detection

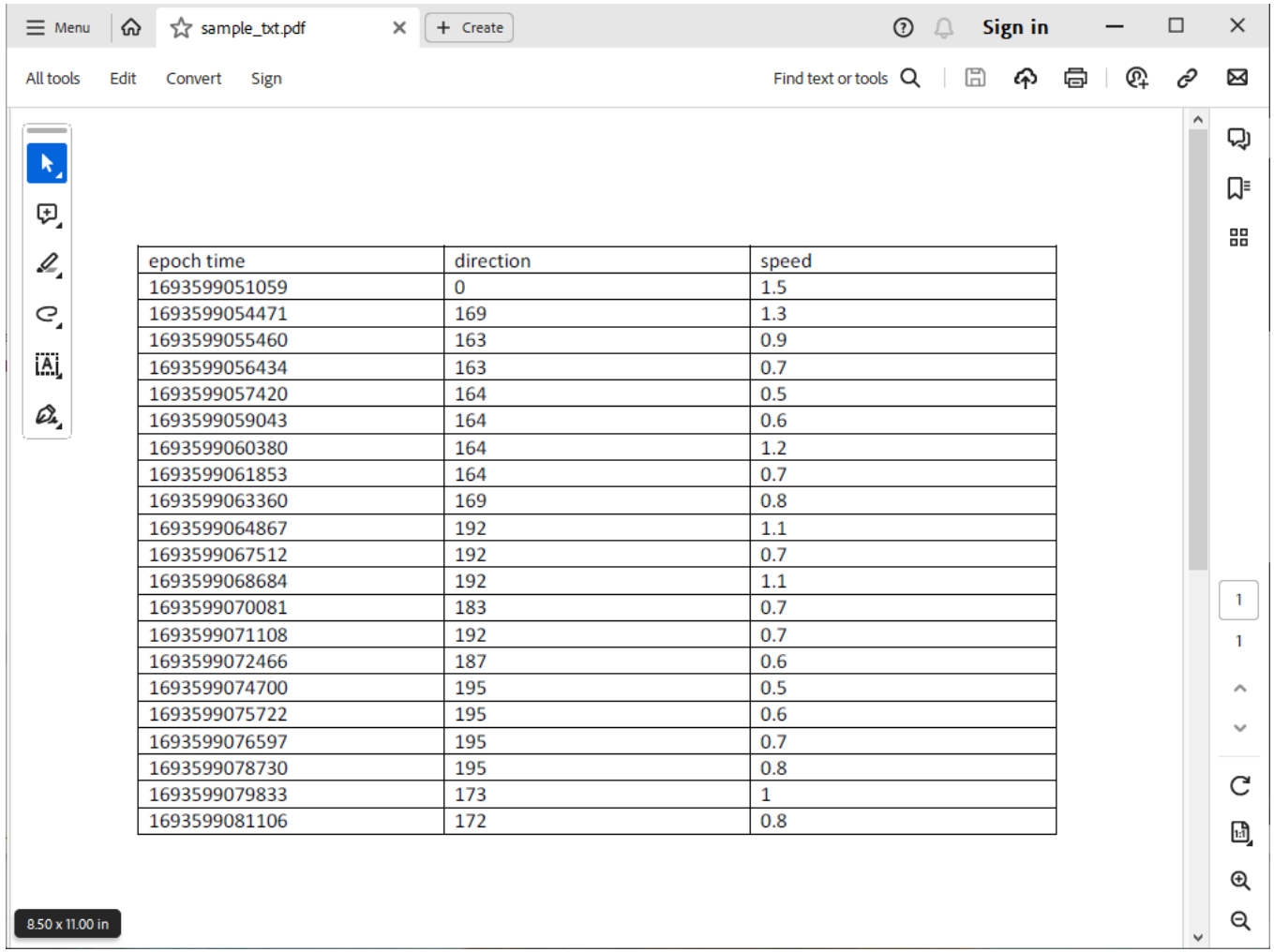
Wind Industrial IoT

Methane Industrial IoT

# Example: Wind IIoT Data

| epoch time | direction | speed |
|---|---|---|
| 1693599051059 | 0 | 1.5 |
| 1693599054471 | 169 | 1.3 |
| 1693599055460 | 163 | 0.9 |
| 1693599056434 | 163 | 0.7 |
| 1693599057420 | 164 | 0.5 |
| 1693599059043 | 164 | 0.6 |
| 1693599060380 | 164 | 1.2 |
| 1693599061853 | 164 | 0.7 |
| 1693599063360 | 169 | 0.8 |
| 1693599064867 | 192 | 1.1 |
| 1693599067512 | 192 | 0.7 |
| 1693599068684 | 192 | 1.1 |
| 1693599070081 | 183 | 0.7 |
| 1693599071108 | 192 | 0.7 |
| 1693599072466 | 187 | 0.6 |
| 1693599074700 | 195 | 0.5 |
| 1693599075722 | 195 | 0.6 |
| 1693599076597 | 195 | 0.7 |
| 1693599078730 | 195 | 0.8 |
| 1693599079833 | 173 | 1 |
| 1693599081106 | 172 | 0.8 |

# Adobe PDF with Texts




| epoch time | direction | speed |
|---|---|---|
| 1693599051059 | 0 | 1.5 |
| 1693599054471 | 169 | 1.3 |
| 1693599055460 | 163 | 0.9 |
| 1693599056434 | 163 | 0.7 |
| 1693599057420 | 164 | 0.5 |
| 1693599059043 | 164 | 0.6 |
| 1693599060380 | 164 | 1.2 |
| 1693599061853 | 164 | 0.7 |
| 1693599063360 | 169 | 0.8 |
| 1693599064867 | 192 | 1.1 |
| 1693599067512 | 192 | 0.7 |
| 1693599068684 | 192 | 1.1 |
| 1693599070081 | 183 | 0.7 |
| 1693599071108 | 192 | 0.7 |
| 1693599072466 | 187 | 0.6 |
| 1693599074700 | 195 | 0.5 |
| 1693599075722 | 195 | 0.6 |
| 1693599076597 | 195 | 0.7 |
| 1693599078730 | 195 | 0.8 |
| 1693599079833 | 173 | 1 |
| 1693599081106 | 172 | 0.8 |

# Adobe PDF with Texts

```
ecet@ecetbox:~/sample$ pdftotext sample_txt.pdf
ecet@ecetbox:~/sample$ cat sample_txt.txt
epoch time
1693599051059
1693599054471
1693599055460
1693599056434
1693599057420
1693599059043
1693599060380
1693599061853
1693599063360
1693599064867
1693599067512
1693599068684
1693599070081
1693599071108
1693599072466
1693599074700
1693599075722
1693599076597
1693599078730
1693599079833
1693599081106

direction
0
169
163
163
164
164
164
164
169
192
192
192
183
192
187
195
195
195
195
173
172
```

```
speed
1.5
1.3
0.9
0.7
0.5
0.6
1.2
0.7
0.8
1.1
0.7
1.1
0.7
0.7
0.6
0.5
0.6
0.7
0.8
1
0.8
```

# Adobe PDF with Scanned Image

```
ecet@ecetbox:~/sample$ pdftotext sample_img.pdf
ecet@ecetbox:~/sample$ cat sample_img.txt

ecet@ecetbox:~/sample$
```

- Image PDF
  - pdftotext does not use Optical Character Recognition
  - Output contains No TEXT
- OCR based Tool
  - OCRMYPDF: Integrates Tesseract OCR

# Adobe PDF with Scanned Image

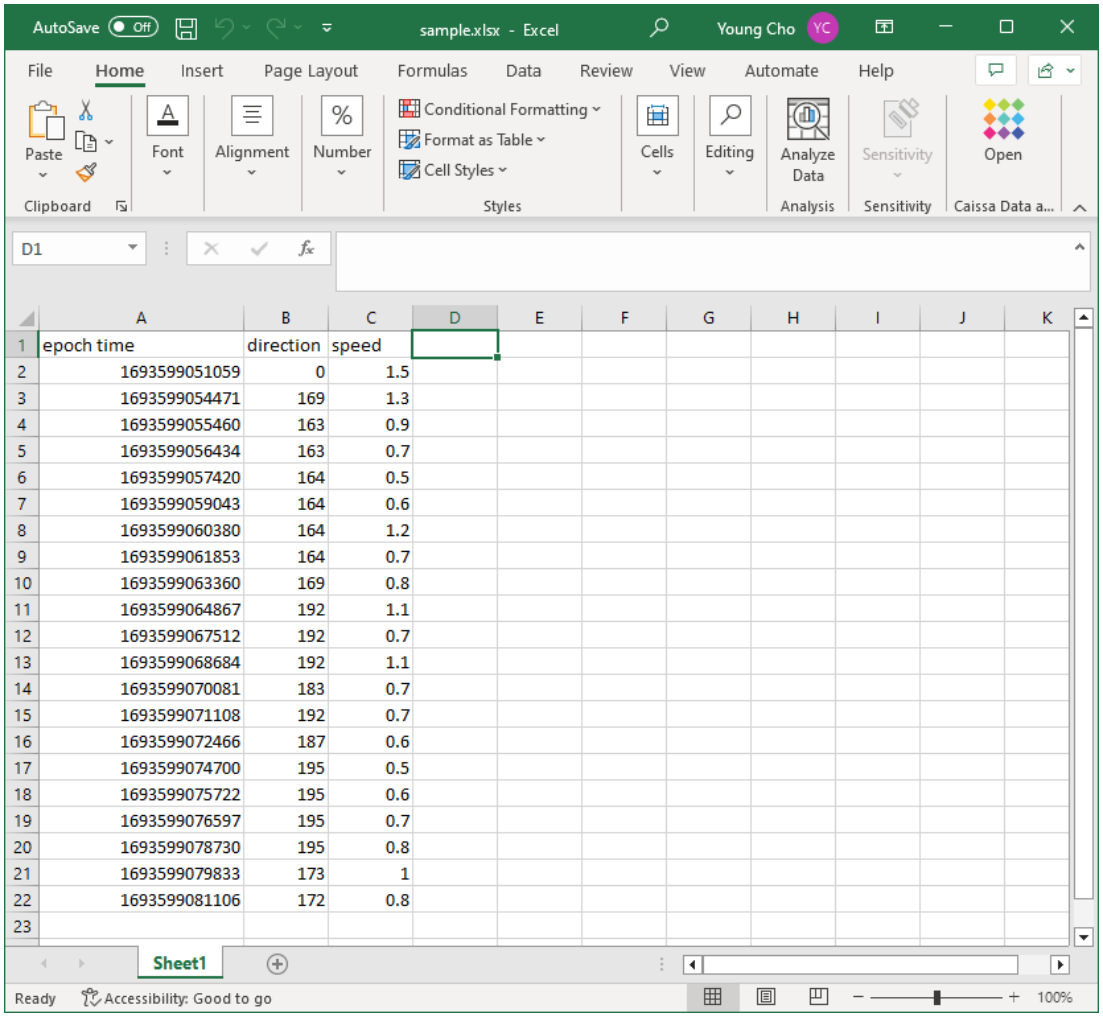
```
ecet@ecetbox:~/sample$ sudo apt install ocrmypdf
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Suggested packages:
  ocrmypdf-doc python-watchdog img2pdf
The following NEW packages will be installed:
  ocrmypdf
0 upgraded, 1 newly installed, 0 to remove and 42 not upgraded.
Need to get 115 kB of archives.
After this operation, 561 kB of additional disk space will be used.
Get:1 http://us.archive.ubuntu.com/ubuntu jammy/universe amd64 ocrmypdf all 13.4.0+dfsg-1 [115 kB]
Fetched 115 kB in 1s (161 kB/s)
Selecting previously unselected package ocrmypdf.
(Reading database ... 288739 files and directories currently installed.)
Preparing to unpack .../ocrmypdf_13.4.0+dfsg-1_all.deb ...
Unpacking ocrmypdf (13.4.0+dfsg-1) ...
Setting up ocrmypdf (13.4.0+dfsg-1) ...
Processing triggers for man-db (2.10.2-1) ...
ecet@ecetbox:~/sample$ ocrmypdf --force-ocr sample_img.pdf sample_ocrout.pdf
Scanning contents: 100%|└────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┘| 1/1 [00:00<00:00, 309.98page/s]
    1 page already has text! - rasterizing text and running OCR anyway
OCR: 100%|└────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┘| 1.0/1.0 [00:04<00:00,  4.76s/page]
Postprocessing...
PDF/A conversion: 100%|└────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┘| 1/1 [00:00<00:00,  1.85page/s]
Recompressing JPEGs: 0image [00:00, ?image/s]
Deflating JPEGs: 100%|└────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┘| 1/1 [00:00<00:00, 95.77image/s]
JBIG2: 0item [00:00, ?item/s]
Optimize ratio: 2.60 savings: 61.6%
Output file is a PDF/A-2B (as expected)
ecet@ecetbox:~/sample$
```

# Adobe PDF with Scanned Image

```
ecet@ecetbox:~/sample$ pdftotext sample_ocrout.pdf
ecet@ecetbox:~/sample$ cat sample_ocrout.txt
epoch time

1693599051059
1693599054471
1693599055460
1693599056434
1693599057420
1693599059043
1693599060380
1693599061853
1693599063360
1693599064867
1693599067512
1693599068684
1693599070081
1693599071108
1693599072466
1693599074700
1693599075722
1693599076597
1693599078730
1693599079833
1693599081106

direction

0
169
163
163
164
164
164
164
169
192
192
192
183
192
187
195
195
195
195
173
172
```

```
speed

1.5
1.3
0.9
0.7
0.5
0.6
1.2
0.7
0.8
1.1
0.7
1.1
0.7
0.7
0.6
0.5
0.6
0.7
0.8
]
0.8
```

# Microsoft Excel (.xlsx)




| | A | B | C |
|---|---|---|---|
| 1 | epoch time | direction | speed |
| 2 | 1693599051059 | 0 | 1.5 |
| 3 | 1693599054471 | 169 | 1.3 |
| 4 | 1693599055460 | 163 | 0.9 |
| 5 | 1693599056434 | 163 | 0.7 |
| 6 | 1693599057420 | 164 | 0.5 |
| 7 | 1693599059043 | 164 | 0.6 |
| 8 | 1693599060380 | 164 | 1.2 |
| 9 | 1693599061853 | 164 | 0.7 |
| 10 | 1693599063360 | 169 | 0.8 |
| 11 | 1693599064867 | 192 | 1.1 |
| 12 | 1693599067512 | 192 | 0.7 |
| 13 | 1693599068684 | 192 | 1.1 |
| 14 | 1693599070081 | 183 | 0.7 |
| 15 | 1693599071108 | 192 | 0.7 |
| 16 | 1693599072466 | 187 | 0.6 |
| 17 | 1693599074700 | 195 | 0.5 |
| 18 | 1693599075722 | 195 | 0.6 |
| 19 | 1693599076597 | 195 | 0.7 |
| 20 | 1693599078730 | 195 | 0.8 |
| 21 | 1693599079833 | 173 | 1 |
| 22 | 1693599081106 | 172 | 0.8 |
| 23 | | | |

# Microsoft Excel

- Microsoft Excel "Save As…"
  - Possible to Convert to TXT or CSV
  - Not automated
- Command Line Tool
  - Much more powerful automation
  - Tools
    - UNZIP (*.xlsx file is ZIP file)
    - CSVKIT (converts *.xlsx into CSV format)

# Microsoft Excel

```
ecet@ecetbox:~/sample/sample_xlsx$ unzip sample.xlsx
Archive:  sample.xlsx
  inflating: [Content_Types].xml
  inflating: _rels/.rels
  inflating: xl/workbook.xml
  inflating: xl/_rels/workbook.xml.rels
  inflating: xl/worksheets/sheet1.xml
  inflating: xl/theme/theme1.xml
  inflating: xl/styles.xml
  inflating: xl/sharedStrings.xml
  inflating: docProps/core.xml
  inflating: docProps/app.xml
ecet@ecetbox:~/sample/sample_xlsx$ more xl/worksheets/sheet1.xml
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<worksheet xmlns="http://schemas.openxmlformats.org/spreadsheetml/2006/main" xmlns:r="http://schemas.openxmlformats.org/officeDocument/2006/relationships" x
mlns:mc="http://schemas.openxmlformats.org/markup-compatibility/2006" mc:Ignorable="x14ac xr xr2 xr3" xmlns:x14ac="http://schemas.microsoft.com/office/sprea
dsheetml/2009/9/ac" xmlns:xr="http://schemas.microsoft.com/office/spreadsheetml/2014/revision" xmlns:xr2="http://schemas.microsoft.com/office/spreadsheetml/
2015/revision2" xmlns:xr3="http://schemas.microsoft.com/office/spreadsheetml/2016/revision3" xr:uid="{BE83200F-09F8-472C-A414-0D743605C97A}"><dimension ref=
"A1:C22"/><sheetViews><sheetView tabSelected="1" workbookViewId="0"><selection activeCell="A12" sqref="A12"/></sheetView></sheetViews><sheetFormatPr default
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.25"><c r="A2" s="1"><v>1693599051059</v></c><c r="B2"><v>0</v></c><c r="C2"><v>1.5</v></c></row><row r="3" spans="1:3" x14ac:dyDescent="0.25"><c r="A3" s="
1"><v>1693599054471</v></c><c r="B3"><v>169</v></c><c r="C3"><v>1.3</v></c></row><row r="4" spans="1:3" x14ac:dyDescent="0.25"><c r="A4" s="1"><v>1693599055
460</v></c><c r="B4"><v>163</v></c><c r="C4"><v>0.9</v></c></row><row r="5" spans="1:3" x14ac:dyDescent="0.25"><c r="A5" s="1"><v>1693599056434</v></c><c r=
"B5"><v>163</v></c><c r="C5"><v>0.7</v></c></row><row r="6" spans="1:3" x14ac:dyDescent="0.25"><c r="A6" s="1"><v>1693599057420</v></c><c r="B6"><v>164</v><
/c><c r="C6"><v>0.5</v></c></row><row r="7" spans="1:3" x14ac:dyDescent="0.25"><c r="A7" s="1"><v>1693599059043</v></c><c r="B7"><v>164</v></c><c r="C7"><v>
0.6</v></c></row><row r="8" spans="1:3" x14ac:dyDescent="0.25"><c r="A8" s="1"><v>1693599060380</v></c><c r="B8"><v>164</v></c><c r="C8"><v>1.2</v></c></row
><row r="9" spans="1:3" x14ac:dyDescent="0.25"><c r="A9" s="1"><v>1693599061853</v></c><c r="B9"><v>164</v></c><c r="C9"><v>0.7</v></c></row><row r="10" spa
ns="1:3" x14ac:dyDescent="0.25"><c r="A10" s="1"><v>1693599063360</v></c><c r="B10"><v>169</v></c><c r="C10"><v>0.8</v></c></row><row r="11" spans="1:3" x14
ac:dyDescent="0.25"><c r="A11" s="1"><v>1693599064867</v></c><c r="B11"><v>192</v></c><c r="C11"><v>1.1000000000000001</v></c></row><row r="12" spans="1:3"
x14ac:dyDescent="0.25"><c r="A12" s="1"><v>1693599067512</v></c><c r="B12"><v>192</v></c><c r="C12"><v>0.7</v></c></row><row r="13" spans="1:3" x14ac:dyDesc
ent="0.25"><c r="A13" s="1"><v>1693599068684</v></c><c r="B13"><v>192</v></c><c r="C13"><v>1.1000000000000001</v></c></row><row r="14" spans="1:3" x14ac:dyD
escent="0.25"><c r="A14" s="1"><v>1693599070081</v></c><c r="B14"><v>183</v></c><c r="C14"><v>0.7</v></c></row><row r="15" spans="1:3" x14ac:dyDescent="0.25
"><c r="A15" s="1"><v>1693599071108</v></c><c r="B15"><v>192</v></c><c r="C15"><v>0.7</v></c></row><row r="16" spans="1:3" x14ac:dyDescent="0.25"><c r="A16"
 s="1"><v>1693599072466</v></c><c r="B16"><v>187</v></c><c r="C16"><v>0.6</v></c></row><row r="17" spans="1:3" x14ac:dyDescent="0.25"><c r="A17" s="1"><v>16
93599074700</v></c><c r="B17"><v>195</v></c><c r="C17"><v>0.5</v></c></row><row r="18" spans="1:3" x14ac:dyDescent="0.25"><c r="A18" s="1"><v>1693599075722<
/v></c><c r="B18"><v>195</v></c><c r="C18"><v>0.6</v></c></row><row r="19" spans="1:3" x14ac:dyDescent="0.25"><c r="A19" s="1"><v>1693599076597</v></c><c r=
"B19"><v>195</v></c><c r="C19"><v>0.7</v></c></row><row r="20" spans="1:3" x14ac:dyDescent="0.25"><c r="A20" s="1"><v>1693599078730</v></c><c r="B20"><v>195
</v></c><c r="C20"><v>0.8</v></c></row><row r="21" spans="1:3" x14ac:dyDescent="0.25"><c r="A21" s="1"><v>1693599079833</v></c><c r="B21"><v>173</v></c><c r
="C21"><v>1</v></c></row><row r="22" spans="1:3" x14ac:dyDescent="0.25"><c r="A22" s="1"><v>1693599081106</v></c><c r="B22"><v>172</v></c><c r="C22"><v>0.8<
/v></c></row></sheetData><pageMargins left="0.7" right="0.7" top="0.75" bottom="0.75" header="0.3" footer="0.3"/></worksheet>
ecet@ecetbox:~/sample/sample_xlsx$
```
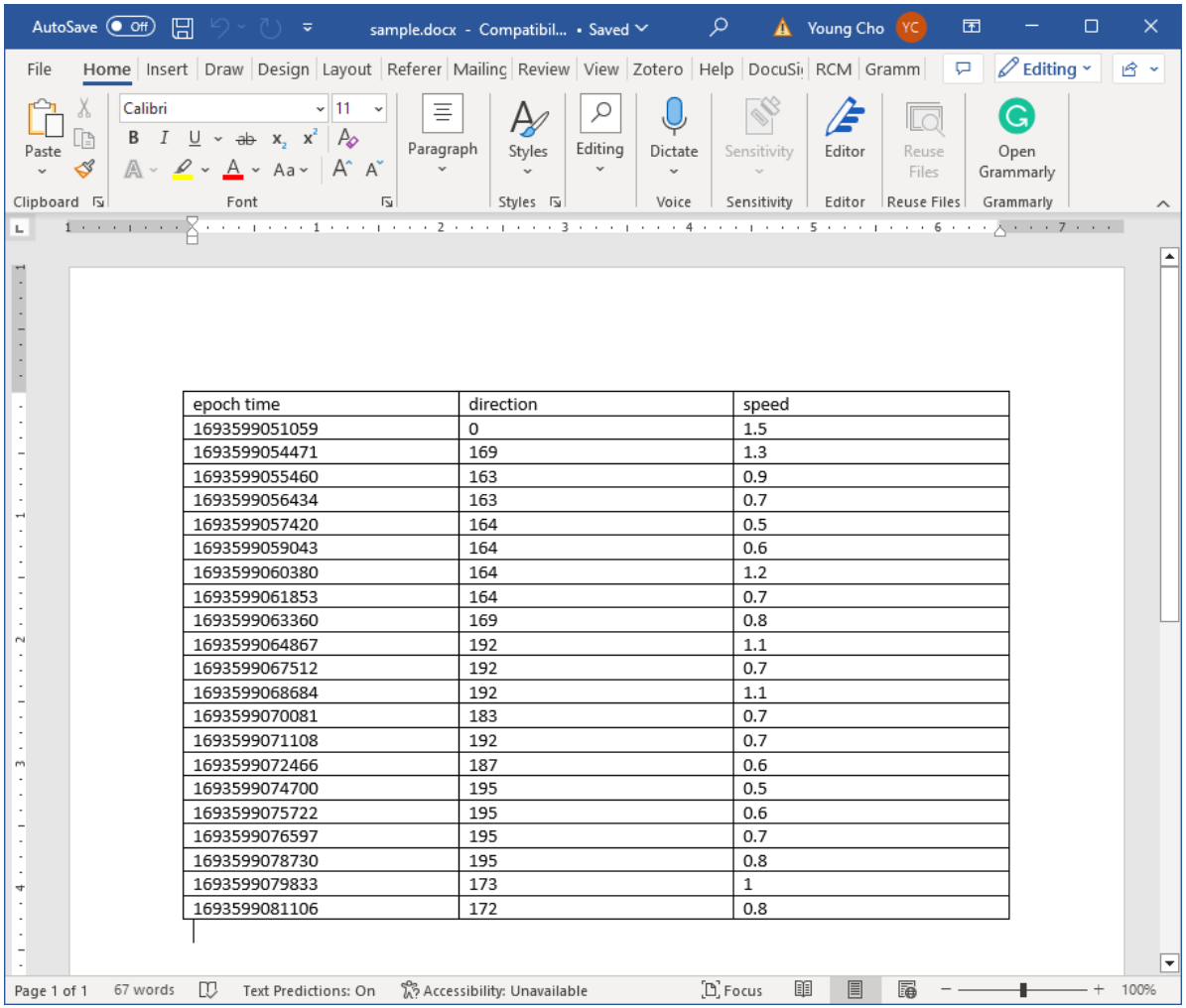

# Microsoft Excel

```
ecet@ecetbox:~/sample/sample_xlsx$ sudo apt install csvkit
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Suggested packages:
  csvkit-doc
The following NEW packages will be installed:
  csvkit
0 upgraded, 1 newly installed, 0 to remove and 42 not upgraded.
Need to get 27.2 kB of archives.
After this operation, 56.3 kB of additional disk space will be used.
Get:1 http://us.archive.ubuntu.com/ubuntu jammy/universe amd64 csvkit all 1.0.6-1 [27.2 kB]
Fetched 27.2 kB in 0s (55.3 kB/s)
Selecting previously unselected package csvkit.
(Reading database ... 288772 files and directories currently installed.)
Preparing to unpack .../csvkit_1.0.6-1_all.deb ...
Unpacking csvkit (1.0.6-1) ...
Setting up csvkit (1.0.6-1) ...
Processing triggers for man-db (2.10.2-1) ...
ecet@ecetbox:~/sample/sample_xlsx$ in2csv sample.xlsx
epoch time,direction ,speed
1693599051059,0,1.5
1693599054471,169,1.3
1693599055460,163,0.9
1693599056434,163,0.7
1693599057420,164,0.5
1693599059043,164,0.6
1693599060380,164,1.2
1693599061853,164,0.7
1693599063360,169,0.8
1693599064867,192,1.1
1693599067512,192,0.7
1693599068684,192,1.1
1693599070081,183,0.7
1693599071108,192,0.7
1693599072466,187,0.6
1693599074700,195,0.5
1693599075722,195,0.6
1693599076597,195,0.7
1693599078730,195,0.8
1693599079833,173,1
1693599081106,172,0.8
ecet@ecetbox:~/sample/sample_xlsx$
```

# Microsoft Word (*.doc/*.docx)




| epoch time | direction | speed |
|---|---|---|
| 1693599051059 | 0 | 1.5 |
| 1693599054471 | 169 | 1.3 |
| 1693599055460 | 163 | 0.9 |
| 1693599056434 | 163 | 0.7 |
| 1693599057420 | 164 | 0.5 |
| 1693599059043 | 164 | 0.6 |
| 1693599060380 | 164 | 1.2 |
| 1693599061853 | 164 | 0.7 |
| 1693599063360 | 169 | 0.8 |
| 1693599064867 | 192 | 1.1 |
| 1693599067512 | 192 | 0.7 |
| 1693599068684 | 192 | 1.1 |
| 1693599070081 | 183 | 0.7 |
| 1693599071108 | 192 | 0.7 |
| 1693599072466 | 187 | 0.6 |
| 1693599074700 | 195 | 0.5 |
| 1693599075722 | 195 | 0.6 |
| 1693599076597 | 195 | 0.7 |
| 1693599078730 | 195 | 0.8 |
| 1693599079833 | 173 | 1 |
| 1693599081106 | 172 | 0.8 |

# Microsoft Word 97-2003 (*.doc)

```
ecet@ecetbox:~/sample$ sudo apt install catdoc
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Suggested packages:
  tk | wish
The following NEW packages will be installed:
  catdoc
0 upgraded, 1 newly installed, 0 to remove and 42 not upgraded.
Need to get 89.5 kB of archives.
After this operation, 695 kB of additional disk space will be used.
Get:1 http://us.archive.ubuntu.com/ubuntu jammy/universe amd64 catdoc amd64 1:0.95-5 [89.5 kB]
Fetched 89.5 kB in 1s (123 kB/s)
Selecting previously unselected package catdoc.
(Reading database ... 288727 files and directories currently installed.)
Preparing to unpack .../catdoc_1%3a0.95-5_amd64.deb ...
Unpacking catdoc (1:0.95-5) ...
Setting up catdoc (1:0.95-5) ...
Processing triggers for mailcap (3.70+nmu1ubuntu1) ...
Processing triggers for man-db (2.10.2-1) ...
```

# Microsoft Word 97-2003 (*.doc)

```
ecet@ecetbox:~/sample$ catdoc sample.doc > sample_doc.txt
ecet@ecetbox:~/sample$ more sample_doc.txt
epoch time      direction       speed

1693599051059   0       1.5

1693599054471   169     1.3

1693599055460   163     0.9

1693599056434   163     0.7

1693599057420   164     0.5

1693599059043   164     0.6

1693599060380   164     1.2

1693599061853   164     0.7

1693599063360   169     0.8

1693599064867   192     1.1

1693599067512   192     0.7

1693599068684   192     1.1

1693599070081   183     0.7

1693599071108   192     0.7

1693599072466   187     0.6

1693599074700   195     0.5

1693599075722   195     0.6

1693599076597   195     0.7

1693599078730   195     0.8

1693599079833   173     1

1693599081106   172     0.8
```

# Microsoft Word (*.docx)

```
ecet@ecetbox:~/sample$ sudo apt install docx2txt
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  docx2txt
0 upgraded, 1 newly installed, 0 to remove and 42 not upgraded.
Need to get 15.7 kB of archives.
After this operation, 47.1 kB of additional disk space will be used.
Get:1 http://us.archive.ubuntu.com/ubuntu jammy/universe amd64 docx2txt all 1.4-5 [15.7 kB]
Fetched 15.7 kB in 0s (34.0 kB/s)
Selecting previously unselected package docx2txt.
(Reading database ... 288795 files and directories currently installed.)
Preparing to unpack .../docx2txt_1.4-5_all.deb ...
Unpacking docx2txt (1.4-5) ...
Setting up docx2txt (1.4-5) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for mailcap (3.70+nmu1ubuntu1) ...
ecet@ecetbox:~/sample$ docx2txt sample.docx
```

# Microsoft Word (*.docx)

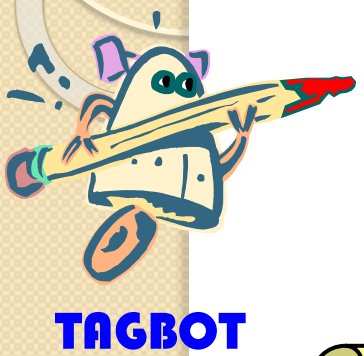
```
ecet@ecetbox:~/sample$ cat sample.txt
epoch time
direction
speed
1693599051059
0
1.5
1693599054471
169
1.3
1693599055460
163
0.9
1693599056434
163
0.7
1693599057420
164
0.5
1693599059043
164
0.6
1693599060380
164
1.2
1693599061853
164
0.7
1693599063360
169
0.8
1693599064867
192
1.1
1693599067512
192
0.7
```

```
1693599068684
192
1.1
1693599070081
183
0.7
1693599071108
192
0.7
1693599072466
187
0.6
1693599074700
195
0.5
1693599075722
195
0.6
1693599076597
195
0.7
1693599078730
195
0.8
1693599079833
173
1
1693599081106
172
0.8
```

# Example: Internet Documents

- Example Project: Application Level Processor (2005-2008)
- Synergistic Effort between Gov. Agencies, Universities, and Industry
  - NSA and Air Force
  - SAIC and Aerospace Corporation
  - Washington University in St. Louis
- Data Quantity and Quality
  - Quantity needed to be intelligently reduced
  - Quality needed to be improved in terms of relevance
- Approach
  - Real-time Stream Data Preparation
  - Context Free Grammar based data cleanser
  - Hardware Accelerated system

# Using Pattern Matchers

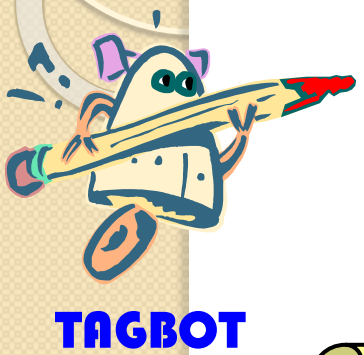



*HTML Source Document*

```
<h2>Company Overview</h2>
<p> Corporate Fact Sheet <br>Founded by <a href=
"/about/profile.html">Dr. J. Robert Beyster</a> and a
small group of scientists in 1969, SAIC, a Fortune 500
company, now ranks … and have more than 43,000 <br> Also
update employee number on: saic.com/news/0722.html <br>
employees with offices in over 150 cities. <br>
```

- String Patterns
  - HTML tags can be detected and marked
  - Marks can be used to filter out the tags
  - Discrete gates, Memory based, Hybrid filter, Bloom filter and etc.
- Detect and Filter out HTML tags
  - <h2>, </h2>, , , <a href=, </a>, <!--, -->
- Some unwanted texts are still not filtered away!

# Beyond Pattern Matching




*HTML Source Document*

```
hdr2                strg                hdr2
<h2>Company Overview</h2>
                    comm                para                strg                link    href
<!-- Corporate Fact Sheet --> <p>Founded by <a href=
                quot                strg                link
"/about/profile.html">Dr. J. Robert Beyster</a> and a
small group of scientists in 1969, SAIC, a Fortune 500
company, now ranks … and have more than 43,000 <!- Also
                                    comm
update employee number on: saic.com/news/0722.html -->
                    strg                                para
employees with offices in over 150 cities. </p>
```

## Token List

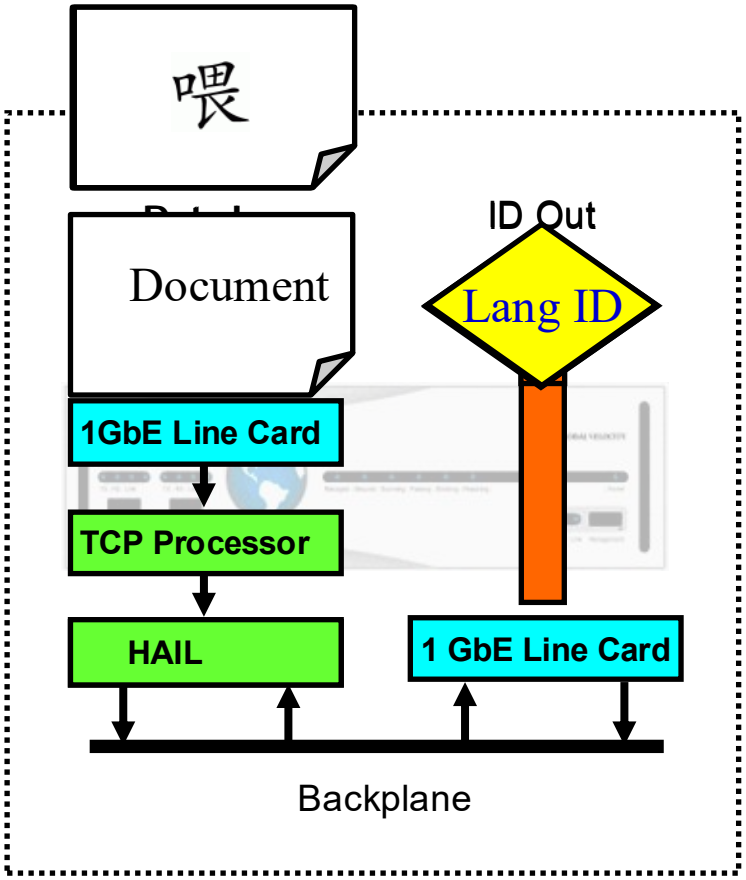
(1) hdr2 : 'h2'
(2) para : 'p'
(3) link : 'a'
(4) href : 'href='
(5) quot : '"" .alpnum*. '""'
(6) comm : alpnum*
(7) strg : alpnum*

## Simple HTML Grammar

(1) Tag_Name → hdr2 | para | link
(2) Comment → '<!-'.comm.'-->'
(3) Attrib → href.quot | ϵ
(4) Tag_Head → '<'.Tag_Name.Attrib.'>'
(5) Tag_Tail → '</'.Tag_Name.'>'
(6) Expr → Comment | strg | ϵ
(7) Line → Tag_Head.Line.Tag_Tail
| Expr.Line.Expr | Expr
(8) Content → Line.Content

# Sensing Languages

- HAIL
  - Above 90% Accuracy with Large Raw Documents
  - Can ID each packet
  - 2.4 Gigabits/second
- Platform
  - HW Compute Accelerators
  - Modular and Stackable Cards
  - Reconfigurable Devices
- Problem
  - Input NOT clean
  - Packets can be small
  - HTML tags and E-mail headers
  - Variety of other attributes



# Processing E-mail

Remove E-mail Headers and HTML Tags

Convert Text

```
From sender@bang.smtp.server.com Fri Jul  7 13:16:33 2006 -0500
Return-Path: <sender@smtp.server.com>
X-Original-To: sender@smtp.server.com
Delivered-To: sender@smtp.server.com
```


```
ICi0ur/lPb+sx9W0ur26KSDAzLehv+4gxq/GxL/4ID0gx9G8urfEIMCvv6PB1sDnILrPx9G068el
us4gwve8rrTru+e0wiA3wM8ox/bB9r3DsKMpICK5zLG5wMwguLbEq7/AILnmxNq1qMW4vsa9w77G
KEJEQSkgwLrH4MDHCrW/sOHA2rHdwLsgx6646SA2wNrIuLTjv6EgurmxzcfSILz2IMDWwLsgsM0i
wMy287DtILi7x9+02S4KCsfRIML3vK6067vntMIgwMyzryC/rMfVtLq9ur/NIMD8yK3F68itv6G8
rSAiucyxucDMIL/suK6/zSCw+MG4x9IgwMe757ChIMDWtMCzxCwgtscgua7BprimILTryK24psXr
x9ggx64gvPawoSDA1rTAs8SwoSDB37/kx8/B9iC068itwMcKx/y9xMC6IMHfv+THz8H2IL7KtNki
uOm8rSAiuLbEq7/
```


(고뉴스=김성덕 기자) 아프카니스탄 바그람 기지 앞에서 폭탄테러에 숨진 다산 부대고 윤장호 병장의 비보가 전해지면서 해외파병 부대의 안전에 대한 우려가 높아지고 있는 가운데, 다산부대 복무경험이 있는 전역병이 아프카니스탄의 근무여건과 상황을 털어놨다.

지난 2004년 8월부터 2005년 2월 중순까지 다산부대에서 통역병으로 근무한 천영록 씨는 "환경적으로는 거의우리나라 사람은 접해보기 힘들 정도로 열악한 곳"이라며 "바그람 고지는 모래투성이인데, 모래가 우리가 생각하는 것처럼 휘날리는 모래가 아니라 딱딱하고 건조한 땅이고,모래 바람 같은게 한 번 들어오면 바람때문에 얼굴이 아플 정도의 강풍이 이틀의 한 번, 하루의 한 번 세게 불어오고 그랬었다"고 회상했다. Outbound message

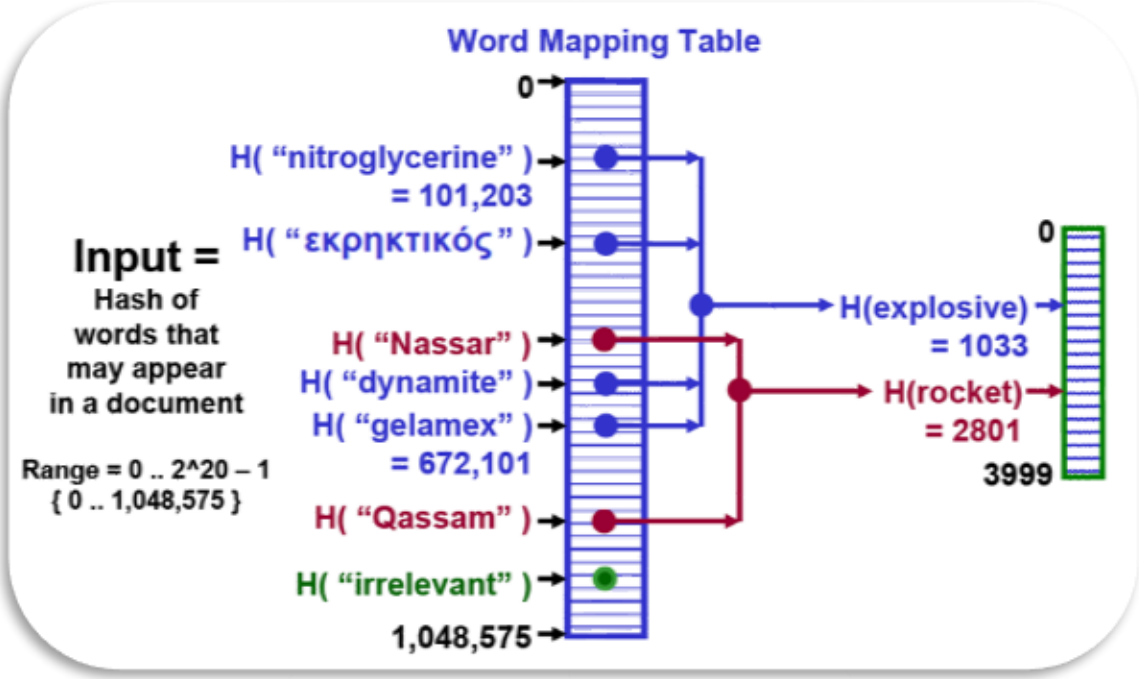
천 씨는 28일 CBS '김현정의 이슈와 사람'과의 인터뷰에서 "(윤병장의 죽음이) 정말 안타까운 일이지만 사실 가 있던 입장에서 보면 그 위험이 그 당시에도 상당히 많았었고, 파병을 갈 때도 상당부분 각오를 하고 가는 부분이기도 했다"며 "위험은 계속 주위에 있었는데, 저희 통제 밖이어서 굉장히 위험한 곳이지만 저희가 어떻게 해볼수 없는, 그냥 각오만 하고 산 경우"라고 당시의 심경을 고백했다!=-4.6 required=6.0 tests=ALL_TRUSTED,AWL,BAYES_00,

```
            HTML_40_50,HTML_MESSAGE autolearn=ham version=3.0.5
X-Original-Status: O
X-Original-X-UID: 3972
Content-Length: 5040
X-Original-X-Keywords:
------=_NextPart_000_000C_01C6A1C7.9102C700
Content-Type: text/html;
            charset="utf-8"
Content-Transfer-Encoding: quoted-printable
```

**Extracting and Conversion of E-mail Body**

# Document Semantics

- Documents transformed to 4000-wide numerical vectors with 4-bit dynamic range
- Document similarity computed based on vector similarity

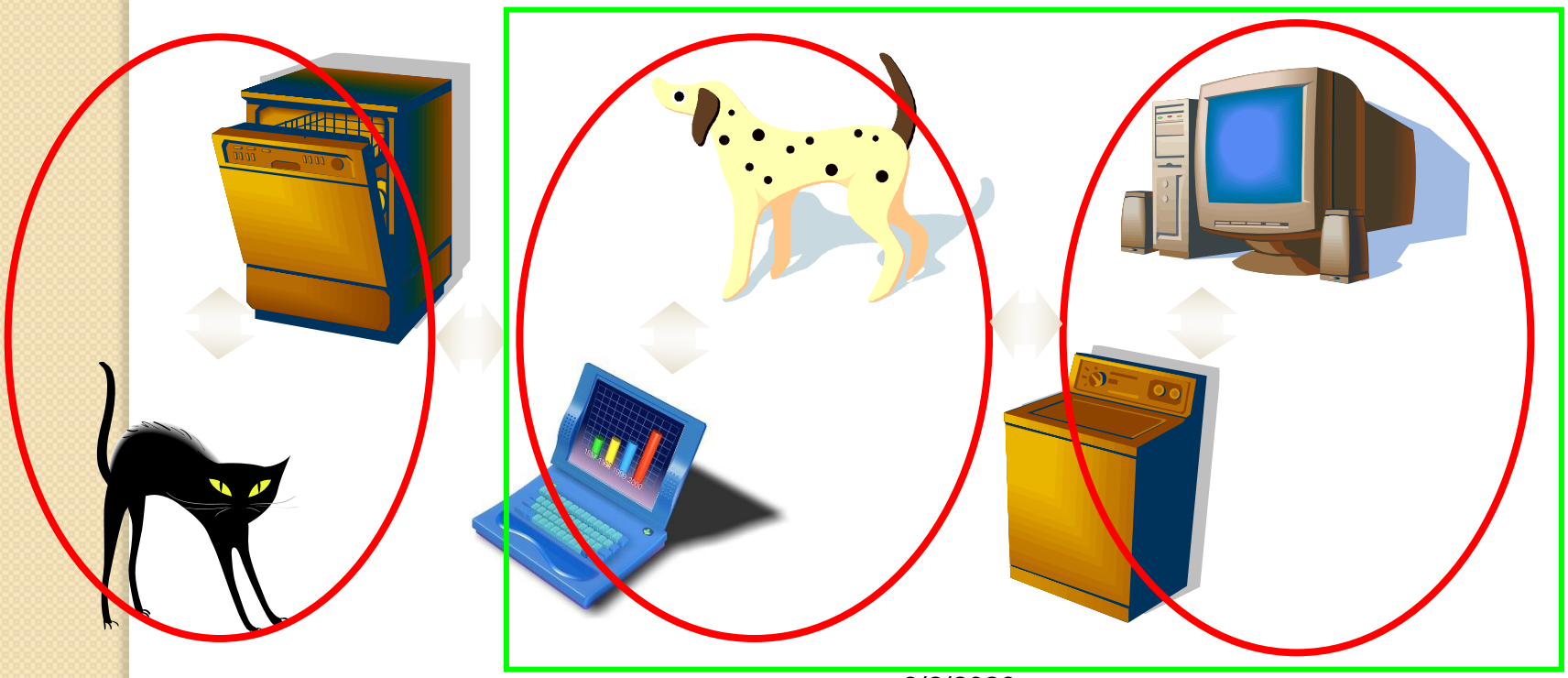


# Finding Knowledge with HW

- Clustering
  - Place similar data points into non-overlapping *clusters*
  - Minimize intra-cluster distances (differences within a cluster)
  - Maximize inter-cluster distances (differences between clusters)

Hierarchical Clustering

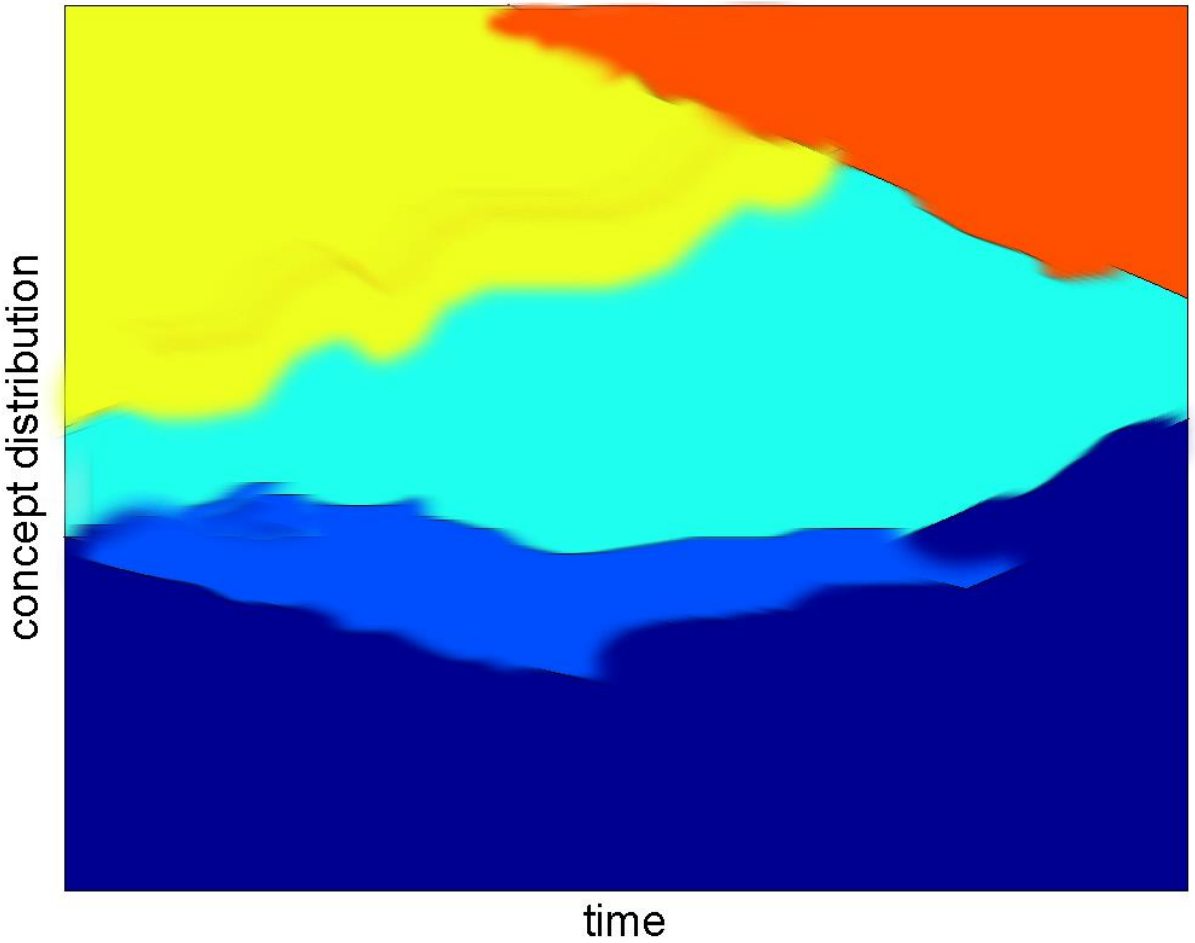
- Cluster the clusters
  - Generates a tree showing hierarchical structure of the data



# Hierarchical Clustering

- Document Insertion
  - Adds a single document to an existing tree
  - Top-down descent, greedy matching
- Batch Clustering
  - Original top-down (global) hierarchical clustering
  - Necessary to avoid getting stuck in local optima
- Document Removal
  1. Dead topics
     - Least Recently Used (LRU) caching (or an approximation)
  2. Over-representation
     - Don't store multiple copies of (nearly) identical documents

# Concept Drift



# Results

- Algorithms trained and tested on CMU 20-newsgroups
  - Standard benchmark (13,000 messages)
  - Added "noise" from talk.origins (11,000 messages)
  - Used K=60, flattened hierarchies to be comparable
- K-means results
  - Vast majority of documents in only two clusters
  - Few concepts discovered
  - Discovered many meaningful concepts
  - However, ~50% of all concepts dominated by noise data
- Streaming Hierarchical Partitioning results
  - Discovered many meaningful concepts
  - Noise data effectively isolated to ~10% of concepts
- **Discovered Concept Vectors leading to Supervised Learning**